Topology Munkres - Solutions Manual

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Contents

Chapter 1

Set Theory and Logic

1.1 Fundamental Concepts

Exercise 1

Okay.

Exercise 2

(a) Only $\Rightarrow$ holds.
(b) Only $\Rightarrow$ holds.
(c) True.
(d) Only $\Leftarrow$ holds.
(e) Only $\subseteq$ holds.
(f) Only $\supseteq$ holds.
(g) True:

$$\begin{gathered} A \cap (B - C) \\ = A \cap (B \cap C^c) \\ = (A \cap B) \cap C^c \\ = \varnothing \cup ((A \cap B) \cap C^c) \\ = ((A \cap B) \cap A^c) \cup ((A \cap B) \cap C^c) \\ = (A \cap B) \cap (A^c \cup C^c) \\ = (A \cap B) \cap (A \cap C)^c \\ = (A \cap B) - (A \cap C) \end{gathered}$$

(h+) Too lazy.

Exercise 3

(a) Contrapositive:
If $x^2 - x \leq 0$, then $x \geq 0$.
Converse:
If $x^2 - x > 0$, then $x < 0$.
Only the contrapositive is true.
(b) Contrapositive:
If $x^2 - x \leq 0$, then $x \leq 0$.
Converse:
If $x^2 - x > 0$, then $x > 0$.
None of them are true.

Exercise 4

(a) There exists $a \in A$ such that $a^2 \notin B$.
(b) For all $a \in A$, we have $a^2 \notin B$.
(c) There exists $a \in A$ such that $a^2 \in B$.
(d) For all $a \notin A$, we have $a^2 \notin B$.

Exercise 5

(a) Both statement and converse are true by definition.
(b) Statement is false. Converse is true.
(c) Statement is true. Converse is false.
(d) Both statement and converse are true by definition.

Exercise 6

Too lazy.

Exercise 7

$$D = A \cap (B \cup C)$$

$$E = (A \cap B) \cup C$$

$$F = A - (B - C)$$

Exercise 8

If A has n elements, then $\mathcal{P}(A)$ has 2^n elements.

Exercise 9

Done this so many times. In other textbooks.

Exercise 10

(a) Yes. $\mathbb{Z} \times \mathbb{R}$
(b) Yes. $\mathbb{R} \times (0, 1]$
(c) No.
(d) Yes. $(\mathbb{R} - \mathbb{Z}) \times \mathbb{Z}$
(e) No.

1.2 Functions

Exercise 1

(a) For any function, we have

$$x \in A_0 \quad \Rightarrow f(x) \in f(A_0) \quad \Rightarrow x \in f^{-1}(f(A_0))$$

Now, let f injective. By definition,

$$x \in f^{-1}(f(A_0)) \quad \Rightarrow f(x) \in f(A_0)$$

Suppose for contradiction that $x \notin A_0$. Denote $y = f(x)$. We know y is an element of $f(A_0)$. In other words, some element of A_0 must be able to create it, i.e. $y = f(z)$ for some $z \in A_0$. As we said, since it's not x, then there must exist some other $z \in A_0$ where $z \neq x$, such that $f(z) = y$. However, this contradicts the fact that f is injective, i.e. we have $z \neq x \Rightarrow f(z) = f(x)$. Hence, $x \in A_0$ is true.
(b) If $y \in f(f^{-1}(B_0))$, then $y = f(x)$ for some $x \in f^{-1}(B_0)$. By definition, the latter means that $f(x) \in B_0$. We know $y = f(x)$, hence $y \in B_0$.

Now, let f be surjective. If $y \in B_0$, then by surjectivity, we have that $f^{-1}(\{y\})$ is non-empty. And by definition, for every element $x \in f^{-1}(\{y\})$, we have $f(x) = y \in B_0$. Hence, $\{y\} \subseteq f(f^{-1}(\{y\}))$, thus $B_0 \subseteq f(f^{-1}(B_0))$.

Exercise 2

(a)
$$x \in f^{-1}(B_0) \quad \Rightarrow f(x) \in B_0 \quad \Rightarrow f(x) \in B_1 \quad \Rightarrow x \in f^{-1}(B_1)$$

(The second "$\Rightarrow$" above uses the assumption given.)
(b)
$$x \in f^{-1}(B_0 \cup B_1) \quad \Rightarrow f(x) \in B_0 \cup B_1$$

If $f(x) \in B_0$, then $x \in f^{-1}(B_0)$. Likewise, if $f(x) \in B_1$, then $x \in f^{-1}(B_1)$. Hence, $x \in f^{-1}(B_0) \cup f^{-1}(B_1)$.

For the reverse direction, suppose that

$$x \in f^{-1}(B_0) \cup f^{-1}(B_1)$$

If $x \in f^{-1}(B_0)$, then $x \in f^{-1}(B_0 \cup B_1)$. Likewise, if $x \in f^{-1}(B_1)$, then $x \in f^{-1}(B_0 \cup B_1)$. Hence, $x \in f^{-1}(B_0 \cup B_1)$.

(c)

$$\begin{gathered}x \in f^{-1}(B_0 \cap B_1) \\ \Leftrightarrow f(x) \in B_0 \cap B_1 \\ \Leftrightarrow f(x) \in B_0 \text{ and } f(x) \in B_1 \\ \Leftrightarrow x \in f^{-1}(B_0) \text{ and } x \in f^{-1}(B_1) \\ \Leftrightarrow x \in f^{-1}(B_0) \cap f^{-1}(B_1)\end{gathered}$$

(d)

$$\begin{gathered}x \in f^{-1}(B_0 - B_1) \\ \Leftrightarrow x \in f^{-1}(B_0 \cap B_1^c) \\ \Leftrightarrow f(x) \in B_0 \cap B_1^c \\ \Leftrightarrow f(x) \in B_0 \text{ and } f(x) \notin B_1 \\ \Leftrightarrow x \in f^{-1}(B_0) \text{ and } x \notin f^{-1}(B_1) \\ \Leftrightarrow x \in f^{-1}(B_0) - f^{-1}(B_1)\end{gathered}$$

(e)

$$y \in f(A_0) \quad \Rightarrow y = f(x) \mid x \in A_0 \quad \Rightarrow y = f(x) \mid x \in A_1 \quad \Rightarrow y \in f(A_1)$$

(The second "$\Rightarrow$" above uses the assumption given.)
(f)

$$y \in f(A_0 \cup A_1) \quad \Rightarrow y = f(x) \mid x \in A_0 \cup A_1$$

If $x \in A_0$, then $y \in f(A_0)$. Otherwise, $y \in f(A_1)$. Hence, $y \in f(A_0) \cup f(A_1)$.
(g)

$$\begin{gathered}y \in f(A_0 \cap A_1) \\ \Rightarrow y = f(x) \mid x \in A_0 \cap A_1 \\ \Rightarrow y = f(x) \mid x \in A_0 \text{ and } y = f(x) \mid x \in A_1 \\ \Rightarrow y \in f(A_0) \text{ and } y \in f(A_1) \\ \Rightarrow y \in f(A_0) \cap f(A_1)\end{gathered}$$

Now, suppose that f is injective.

$$y \in f(A_0) \cap f(A_1)$$

$$\Rightarrow y \in f(A_0) \text{ and } y \in f(A_1)$$

$$\Rightarrow y = f(x_0) \mid x_0 \in A_0 \text{ and } y = f(x_1) \mid x_1 \in A_1$$

By injectivity, $x_0 = x_1$, thus

$$\Rightarrow y = f(x) \mid x \in A_0 \text{ and } x \in A_0$$

$$\Rightarrow y = f(x) \mid x \in A_0 \cap A_1$$

$$\Rightarrow y \in f(A_0 \cap A_1)$$

(h)

$$y \in f(A_0) - f(A_1)$$

$$\Rightarrow y \in f(A_0) \cap f^c(A_1)$$

$$\Rightarrow y \in f(A_0) \text{ and } y \notin f(A_1)$$

$$\Rightarrow y = f(x_0) \mid x_0 \in A_0 \text{ and } y \neq f(x_1) \mid \forall x_1 \in A_1$$

Since $y \in$ range $f|A_0$, there must exist $x \in A_0$ but $x \notin A_1$ such that $f(x) = y$.

$$\Rightarrow y = f(x) \mid x \in A_0 - A_1$$

$$\Rightarrow y \in f(A_0 - A_1)$$

Now, suppose that f is injective.

$$y \in f(A_0 - A_1)$$

$$\Rightarrow y = f(x) \mid x \in A_0 \cap A_1^c$$

We already know, by conjunction elimination, that $y = f(x) \mid x \in A_0$, hence $y \in f(A_0)$.

Now, for the second part, since $x \in A_1^c$, then $x \notin A_1$. Suppose for contradiction that $y \in f(A_1)$. Since $y \in f(A_1)$, there must exist $x' \in A_1$ such that $f(x') = y$. We know $x' \in A_1$ and $x \notin A_1$, hence $x' \neq x$. However, $f(x') = y = f(x)$. This contradicts the injectivity of f. Hence, it must be that $y \notin f(A_1)$.

Thus, $y \in f(A_0)$ and $y \notin f(A_1)$. Hence,

$$\Rightarrow y \in f(A_0) - f(A_1)$$

If you didn't understand, read the following. The crux to understanding the second part is to realize that y is a value in the codomain. Many x can produce this single y. We need to verify that every single $x \in f^{-1}(\{y\})$ is not contained in A_1. If even a single one is found to be in A_1, then the entire y is contained in $f(A_1)$. That's where injectivity comes in, guaranteeing that the preimage is a singleton, so that, as soon as we find a particular $x \notin A_1$ such that $f(x) = y$, then there is no more x to verify.

Exercise 3

Hell nah.

Exercise 4

(a)

$$
\begin{aligned}
(g \circ f)^{-1}(C_0) &= \{a \in A \mid (g \circ f)(a) \in C_0\} \\
&= \{a \in A \mid \exists b \in B : f(a) = b \text{ and } g(b) \in C_0\} \\
&= \{a \in A \mid \exists b \in B : f(a) = b \text{ and } b \in g^{-1}(C_0)\} \\
&= \{a \in A \mid f(a) \in g^{-1}(C_0)\} \\
&= f^{-1}(g^{-1}(C_0))
\end{aligned}
$$

(b)

$$x \neq x' \quad \Rightarrow f(x) \neq f(x') \quad \Rightarrow g(f(x)) \neq g(f(x'))$$

(c) Recall that Munkres uses the word "range" to mean "codomain". Hence, we only require f to be injective. g must be injective on $f(A) \subseteq B$. But it can behave freely elsewhere, i.e. on $B - f(A)$, because these values are never reached by f.
(d) For every $c \in C$, we can find $b \in B$ such that $g(b) = c$. And for every $b \in B$, we can find $a \in A$ such that $f(a) = b$. Thus, for every $c \in C$, we can find $a \in A$ such that $c = g(f(a))$. Hence, $g \circ f$ is surjective.
(e) g must be surjective. However, f need not be surjective, because it's possible that $C = g(B_0)$ for some subset $B_0 \subset B$. In other words, f only needs to reach B_0 for the entire composite to be surjective.
(f) Too lazy.

Exercise 5

(a) Let $f : A \to B$ have a left inverse $g : B \to A$ where $g \circ f = i_A$. Suppose for contradiction that f is not injective, i.e. there exists $x \neq x'$ such that $f(x) = f(x')$. Then,

$$g(f(x)) = g(f(x'))$$

But $g \circ f$ is the identity function, hence

$$\Rightarrow x = x'$$

contradicting our assumption. Hence, f is injective.

Now, let $f : A \to B$ have a right inverse $h : B \to A$ where $f \circ h = i_B$. For every $b \in B$, we can compute $h(b) \in A$. The identity function tells us that $f(h(b)) = b$. In other words, for every $b \in B$, we found $a \in A$ such that $f(a) = b$, where $a = h(b)$. This is precisely the requirement of surjectivity of f.
(b, c) It's really just playing with the codomain.
(d) Yes. Again, when f is not surjective, f need not map to the entirety of B, so g is free to do whatever it likes on the remaining portions. And when f is not injective, it can have multiple right inverses. For example, if $x, x' \in A$ both map to $y \in B$, then $h(b)$ can choose between them.
(e) Well, since f is both injective and surjective, then it's bijective.

First, h is surjective. Suppose for contradiction that it's not surjective. Then take any $x \in A - h(B)$, which is non-empty by non-surjectivity of h. Every $y \in B$ has already been taken by some $x' \in h(B)$, because $f(h(B)) = B$ by property of identity. Since x is in the domain of f, then it must produce something under f, say $y \in B$. But then $f(x) = y = f(x')$. We know $x' \neq x$, because they belong in complementary subsets of A. This would contradict the injectivity of f, hence h is surjective.

Now we can complete the proof. Given $y \in B$, using the surjectivity of f, there exists $x \in A$ such that

$$y = f(x)$$

Hence,

$$g(y) = g(f(x)) = x$$

by property of identity. Now, since h is surjective, then there must exist some $y' \in B$ such that

$$h(y') = x$$

It follows that

$$f(h(y')) = f(x)$$

$$\Rightarrow y' = y$$

Hence, $h(y) = h(y') = x = g(y)$ for all $y \in B$, and since they share this same domain, then they are identical functions.

By injectivity of f, the x that gives $y = f(x)$ is unique. Hence, $g = h = f^{-1}$ is the inverse function, since it evaluates to this unique x.

Exercise 6

Too lazy.

1.3 Relations

Exercise 1

Equality is reflexive, symmetric, and transitive. The equivalence classes can be seen if we fix $y - x^2 = c$. Which becomes $y = x^2 + c$. These are parabolas. The equivalence classes differ by the amount shifted vertically.

Exercise 2

We are basically removing elements from an already related set, and we have to prove that the remaining connections still form an equivalence relation.

For every element $a \in A$, we have $(a, a) \in C$. If $a \in A_0$, then $(a, a) \in A_0 \times A_0$, hence $(a, a) \in C \cap (A_0 \times A_0)$.

Suppose $(a, b) \in C \cap (A_0 \times A_0)$. Clearly, $(b, a) \in C$. And since $a \in A_0$ and $b \in A_0$, then $(b, a) \in A_0 \times A_0$. Thus, $(b, a) \in C \cap (A_0 \times A_0)$.

Suppose $(a, b), (b, c) \in C \cap (A_0 \times A_0)$. Clearly, $(a, c) \in C$. And since $a \in A_0$ and $c \in A_0$, then $(a, c) \in A_0$. Thus, $(a, c) \in C \cap (A_0 \times A_0)$.

Exercise 3

These are conditional statements. If no pair (a, b) exist, then we can't proceed. In other words, none of these statements enforce existence.

Exercise 4

(a) Apply the properties of equality.
(b) Clearly intuitive, but I'm confusing myself with proofs.

Exercise 5

(a) $x - x = 0 \in \mathbb{Z}$, so it's reflexive. If $x - y = n \in \mathbb{Z}$, then $y - x = -n \in \mathbb{Z}$, so it's symmetric. If $x - y = n \in \mathbb{Z}$ and $y - z = m \in \mathbb{Z}$, then $x - z = x + (y - y) - z = (x - y) + (y - z) = n + m \in \mathbb{Z}$, so it's transitive.

Obviously, it's a superset of S, because

$$(x, y) \in S \quad \Rightarrow y = x + 1 \quad \Rightarrow y - x = 1 \in \mathbb{Z} \quad \Rightarrow (x, y) \in S'$$

Numbers belonging to a same equivalence classes have the same mod 1.
(b) If $a \in A$, then (a, a) is contained in every equivalence relation (because that's a property of equivalence relations). Hence, it's in their intersection.

If (a, b) is contained in the intersection, then it must be contained in each equivalence relation. And each equivalence relation must contain (b, a), hence their intersection contains (b, a) as well.

If (a, b) and (b, c) is contained in the intersection, then they must be contained in each equivalence relation. And each equivalence relation must contain (a, c), hence their intersection contains (a, c) as well.
(c) $(x, x) \in T$ for every $x \in \mathbb{R}$. By the way, S is not an equivalence relation, because it's not symmetric, so since $(x, y) \in S$, then $(x, y) \in T$, and since T is an equivalence relation, then $(y, x) \in T$. In other words, we add the missing symmetry. Transitivity doesn't need to be dealt, because the range $0 < x < 2$ is too small for transitivity.

T basically looks like the real number line, but we glue the interval $(0, 1)$ with $(1, 2)$ sort of in a superposition fashion. There's a hole in the middle, with only $(1, 1) \in T$ (not interval, here I'm expressing a pair).

Exercise 6

These are parabolas. Higher parabolas are greater than lower parabolas, and within the same parabola, right points are greater than left points.

First, obviously, every pair of points are comparable, by definition of the relation. Second, if two points are identical, then $y_0 - x_0^2 \neq y_1 - x_1^2$ is not satisfied, then $x_0 \neq x_1$ is not satisfied, hence identical points are not comparable.

Third, transitivity is obvious, but tedious. If $(x_0, y_0) < (x_1, y_1) < (x_2, y_2)$, then the second point either sits on the same or higher parabola than the first, and the third either sits on the same or higher parabola than the second. In the same-same case, we have $y_2 - x_2^2 = y_1 - x_1^2 = y_0 - x_0^2$, so we skip the parabola inequality, directly resulting in $x_0 < x_1 < x_2$, hence $x_0 < x_2$. In the same-higher case, we get $y_2 - x_2^2 > y_1 - x_1^2 = y_0 - x_0^2$, hence $y_2 - x_2^2 > y_0 - x_0^2$. In the higher-same case, we get $y_2 - x_2^2 = y_1 - x_1^2 > y_0 - x_0^2$, hence $y_2 - x_2^2 > y_0 - x_0^2$. In the higher-higher case, we simply have $y_2 - x_2^2 > y_1 - x_1^2 > y_0 - x_0^2$, hence $y_2 - x_2^2 > y_0 - x_0^2$.

Exercise 7

I'm assuming that the restriction excludes points in A outside the restriction:

$$A \to A \cap B$$

$$C \to C \cap (B \times B)$$

For comparability:

$$x, y \in A \cap B$$

$$\Rightarrow x, y \in A \text{ and } x, y \in B$$

Since C is an order relation on A, then we have $(x, y) \in C$ already. And since $x, y \in B$, then $(x, y) \in B \times B$. Hence,

$$\Rightarrow (x, y) \in C \text{ and } (x, y) \in B \times B$$

$$\Rightarrow (x, y) \in C \cap (B \times B)$$

For nonreflexivity, since $(x, x) \notin C$, then $(x, x) \notin C \cap (B \times B)$ even less. Finally, for transitivity

$$(x, y), (y, z) \in C \cap (B \times B)$$

$$\Rightarrow (x, y), (y, z) \in C \text{ and } (x, y), (y, z) \in B \times B$$

$$\Rightarrow (x, z) \in C \text{ and } x, y, z \in B$$

$$\Rightarrow (x, z) \in C \text{ and } (x, z) \in B \times B$$

$$\Rightarrow (x, z) \in C \cap (B \times B)$$

Exercise 8

Too lazy.

Exercise 9

The procedure is similar to Exercise 6.

Exercise 10

(a) Suppose $0 < x < y < 1$. Then $x^2 < y^2$. Then $1 - x^2 > 1 - y^2$. Then $\frac{1}{1-x^2} < \frac{1}{1-y^2}$. Hence, $\frac{x}{1-x^2} < \frac{y}{1-y^2}$. If both are negative, i.e. $-1 < x < y < 0$, then $x^2 > y^2$, then $1 - x^2 < 1 - y^2$, then $\frac{1}{1-x^2} > \frac{1}{1-y^2}$ since both are negative. Hence, x is scaled by a bigger factor, making it more negative $\frac{x}{1-x^2} < \frac{y}{1-y^2}$. Finally, if $x < 0 < y$, then since the denominators are always positive, they maintain their sign, so $\frac{x}{1-x^2} < 0 < \frac{y}{1-y^2}$.
(b) Too lazy.

Exercise 11

Suppose that b, c are both immediate successors of a. Then we know $a < b$ and $a < c$. If $b \neq c$, then either $b < c$ or $c < b$ (comparability property). If $b < c$, then $a < b < c$, which violates the fact that c is immediate successor of a. The other case is similar. Hence, b and c must be equal. Thus, immediate successors are unique. The procedure is similar for immediate predecessors.

Suppose that a, b are both largest elements. Then $a \leq b$, and $b \leq a$, hence $a = b$. Hence, largest elements are unique. The procedure is similar for smallest elements.

Chapter 2

Topological Spaces and Continuous Functions

2.1 Basis for a Topology

Exercise 1

A is equal to the union of U's. We know that topol. spaces are closed under arbitrary union.

Exercise 2

Too lazy.

Exercise 3

Empty set is included, because $X - \varnothing = X$ which is all of X. The set X itself is included, because $X - X = \varnothing$ is countable (it's finite).

Given a collection W of open sets U, which satisfy $X - U$ is countable, we check:

$$X - \bigcup_{U \in W} U = \bigcap_{U \in W} (X - U)$$

We know each $X - U$ is countable. (Of course, one of them could satisfy $X - U = X$, but then it would have no effect on the intersection.) Obviously, arbitrary intersection of countable sets is countable, hence the complement of the union is finite, hence the union is inside $\mathcal{T}_c$.

Finally, if $X - U_0$ and $X - U_1$ are countable, then $X - U_0 \cap U_1 = (X - U_0) \cup (X - U_1)$ is countable, hence $U_0 \cap U_1 \in \mathcal{T}_c$. If one of them satisfies $X - U = X$, then the result is all of X, so it's still included in $\mathcal{T}_c$. Thus $\mathcal{T}_c$ is a topology.

Exercise 4

(a) $\varnothing, X \in \mathcal{T}_\alpha$ for every $\mathcal{T}_\alpha$ already, so they are also included in their intersection. Given a collection of sets in $\bigcap\{\mathcal{T}_\alpha\}$, each set is also individually inside each $\mathcal{T}_\alpha$. Since the latter contain their union, then their intersection also contains their union. Similarly, given a finite amount of sets in $\bigcap\{\mathcal{T}_\alpha\}$, each set is individually in each $\mathcal{T}_\alpha$.

The union $\bigcup\{\mathcal{T}_\alpha\}$ is generally not a topology though. We can take one open set from some $\mathcal{T}_\alpha$ and another from some other $\mathcal{T}_\beta$, and the union of these open sets would not necessarily be included in either topology, hence would not be included in the union of the two topologies.

(b) The smallest surrounding topology must contain every arbitrary union and finite intersection of open sets from each individual topology. It's the smallest, because it satisfies every condition minimally. The largest contained topology is the intersection between the topologies. It's the largest, because it's obvious.

By construction, it's obvious that they are both unique.

(c) Too lazy.

Exercise 5

The topology generated by $\mathcal{A}$ is formed by taking arbitrary unions of open sets in $\mathcal{A}$. The intersection must contain this topology. Every topology containing $\mathcal{A}$ must contain this topology, because it's the smallest, satisfying the conditions of a topological space minimally. Hence the intersection equals this topology. The same goes if $\mathcal{A}$ is a subbasis, where we take finite intersections alongside arbitrary unions.

Exercise 6 - 7

Too lazy.

Exercise 8

(a) Every open set of the standard topology is the arbitrary union of open intervals. Every point x inside an open interval has a neighborhood $(x-\delta, x+\delta)$ fully contained in the interval. We can find a rational between x and $x-\delta$, as well as another one between x and $x+\delta$. Hence, using these two rationals as bounds for a new interval, it shows that for every point x inside an open set, we can find an interval that contains x which is inside the open set.

(b) The open set $[\sqrt{2}, 2) \in \mathbb{R}_l$ cannot be produced by rational-bounded half-open sets.

2.2 The Subspace Topology

Exercise 1

Every open set of A equals $A \cap U$ for some open $U \in Y$. And every open $U \in Y$ equals $Y \cap V$ for some open $V \in X$. Hence, every open set of A equals $A \cap (Y \cap V)$ for some open $V \in X$. But since $A \subseteq Y$, then $A \cap Y = A$, so every open set is simply expressed as $A \cap V$ for some open $V \in X$. Hence A has the subspace topology from X.

Exercise 2

The subspace topology from $\mathcal{T}'$ is finer than that of $\mathcal{T}$, but not necessarily strictly finer.

As an example, let the topology $\mathcal{T}'$ be the topology on $\mathbb{R}$ generated by the basis:

$$\mathcal{B} = \{(a,b) : a, b \in \mathbb{R}\} \cup \{[0,c) : c \in \mathbb{R}\}$$

which is basically the standard topology but we include a half-open ray starting at zero. It is strictly finer than the standard topology, which is $\mathcal{T}$ in our example. However, taking $Y = [0, \infty)$ as the subset produces two equivalent subspace topologies.

Exercise 3

Too lazy.

Exercise 4

Assuming the product topology on $X \times Y$, then every open set in $X \times Y$ is the arbitrary union of a bunch of $U \times V$ for U, V open in X, Y respectively. We know that

$$f \bigcup (U \times V) = \bigcup f(U \times V)$$

Since $\pi_1(U \times V) = U$ which is open, then the union is also open. Hence, π_1 is open. The procedure is similar for π_2.

Exercise 5

(a) If I understand the problem correctly, $X = X'$ denote the same set, but with different topologies $\mathcal{T}$ and $\mathcal{T}'$. Likewise for $Y = Y'$.

The product topology on $X \times Y$ has basis elements of the form $T \times U$ where $T \in \mathcal{T}$ and $U \in \mathcal{U}$. Since $\mathcal{T} \subseteq \mathcal{T}'$ and $\mathcal{U} \subseteq \mathcal{U}'$, then $T \times U \in \mathcal{T}' \times \mathcal{U}'$. Since every basis of $X \times Y$ is included in $X' \times Y'$, then the latter is finer.

(b) Suppose $T \times U \in X \times Y$ implies $T \times U \in X' \times Y'$. Now, given $T \in \mathcal{T}$, since Y is non-empty, then $T \times Y$ is open in $X \times Y$. Hence, $T \times Y \in X' \times Y'$ open as well. Its projection T is open in $\mathcal{T}'$, hence $\mathcal{T} \subseteq \mathcal{T}'$. The procedure is similar for $\mathcal{U}$ and $\mathcal{U}'$.

Exercise 6

Well, for every point x inside rational rectangle $(a,b) \times (c,d)$, we can find a subset irrational rectangle $(e,f) \times (g,h)$ containing x. The converse is true as well. Hence their topologies are equivalent.

Exercise 7

No. For example, take a line with a hole, and let the subset be the entire portion after the hole. This subset is convex, but you cannot express its left endpoint (because it's literally the hole). I got this solution from online.

Exercise 8

It depends on the direction of the line. Too lazy to try all.

Exercise 9

The dictionary order topology's basis are intervals. Given endpoints $\langle a,b\rangle < \langle c,d\rangle$, any interval in the dictionary topology can be decomposed into

$$\{a\} \times (b,\infty)$$

$$(a,c) \times \mathbb{R}$$

$$\{c\} \times (-\infty,d)$$

which belong to $\mathbb{R}_d \times \mathbb{R}$. Conversely, $U \times V$ are the basis for the product topology, where $U \subseteq \mathbb{R}$ are arbitrary (since it's discrete topology) and $V = (p,q) \subseteq \mathbb{R}$ are open intervals. We can express them using the dictionary basis as follows:

$$\bigcup_{x \in U} (\langle x,p\rangle, \langle x,q\rangle)$$

Basically, we are using a bunch of vertical lines. Hence, the two topologies are equal.

Since discrete topology is strictly finer than the standard topology on $\mathbb{R}$, then the product topology must be at least finer as well. In fact, it's also strictly finer, since lines are invalid in the standard topology on $\mathbb{R}^2$, but valid in the other.

Exercise 10

Too lazy.

2.3 Closed Sets and Limit Points

Exercise 1

First, $\varnothing = X - X$ and $X = X - \varnothing$, so both are included. Second, suppose $X - C_\alpha$ are open for (possibly uncountable) α, then $\bigcup_\alpha (X - C_\alpha) = X - \bigcap_\alpha C_\alpha$, the subtracted part which is an arbitary intersection of elements in $\mathcal{C}$, hence also in $\mathcal{C}$, hence the original thing is in $\mathcal{T}$. Finally, suppose $X - C_1$ and $X - C_2$ are open, then $(X - C_1) \cap (X - C_2) = X - (C_1 \cup C_2)$, the subtracted which is a finite union of elements in $\mathcal{C}$, hence also in $\mathcal{C}$, hence the original thing is in $\mathcal{T}$. Hence, it's a topology.

Exercise 2

Since A is closed in Y, and Y is closed in X, then $A = C \cap Y$ for some C closed in X. Intersection of closed sets in X is closed in X, hence A is closed in X.

Exercise 3

Assuming the product topology. We can decompose the complement as

$$X \times Y - A \times B = ((X - A) \times Y) \cup ((X) \times (Y - B))$$

Since $X - A$ is open, and Y is open, and X is open, and $Y - B$ is open, then the Cartesian product and union are open are well. Since the complement of $A \times B$ is open, then it must be closed itself.

Exercise 4

Too lazy.

Exercise 5